

A Software Development Engineer focused on building robust server-side architectures, implementing scalable application functionalities, optimizing database solutions, and ensuring seamless performance and reliability in modern applications.

SKILLS

Web Development

- Backend: Java, Spring Boot, Spring Security, Google Guice
- APIs & Security: REST APIs, JWT, OAuth2
- Frontend: ReactJS
- Cloud: AWS (EC2, ECS, Lambda, S3, DynamoDB, RDS, VPC, IAM, CloudWatch, Glue, Athena)

Database

- SQL: PostgreSQL, Oracle, MySQL
- NoSQL: Redis, DynamoDB

Coding

Languages: Java, Python, SQL, PL/SQL, TypeScript
DSA: <https://leetcode.com/u/rahulravuri/>
GitHub: <https://github.com/rahulravuri>

DevOps

CI&CD: Git, Docker, Jenkins
Testing: Junit, Mockito

EDUCATION

Bachelor of Technology	Aug 2015 - Jun 2019
In <i>Electronics and Communication Engineering</i> from	
Malla Reddy College Of Engineering and Technology, Hyderabad	
Intermediate	JUL 2013 - APR 2015
10th Class	Aug 2012 - APR 2013

CERTIFICATIONS

• Docker And Kubernetes	Udemy	Jan 2025
• SpringBoot JPA	Udemy	May 2024
• Automation Engineer Practitioner	Capgemini	Mar 2020

WORK EXPERIENCE

Amazon <i>Software Development Engineer [Contractor]</i>	May 2025 - Present
- Built backend services and APIs for an internal procurement platform from scratch using a microservice architecture, enabling scalable order processing and integration with internal systems. - Implemented checksum-based validation mechanisms to ensure data integrity and protection during service communication and data transfers. - Designed and developed an ETL pipeline to transfer application order data into a centralized data lakehouse, enabling analytics and reporting through Amazon QuickSight dashboards. - Configured secure networking using VPC peering to enable controlled connectivity between application services and data infrastructure. - Identified and resolved operational and security risks through internal risk management tooling, ensuring compliance with Amazon production standards. - Monitored system performance and implemented load testing to evaluate TPS capacity, tuning services and infrastructure to handle production traffic efficiently.	
Cognizant <i>Software Engineer Associate</i>	Jan 2024 - May 2025
- Improved backend system efficiency by optimizing REST APIs and database queries, reducing response times by 20%. - Refactored database structures and indexing strategies, enhancing query performance and reducing data redundancy. - Implemented backend enhancements to support new application features and improve system scalability. - Collaborated with frontend teams to optimize ReactJS components and API interactions, improving user experience and reducing load times by 15%.	
TCS <i>Business Process Lead</i>	Jan 2022 - Dec 2023
- Designed and implemented backend solutions using AWS services (S3, Lambda) and Kafka to enable seamless data transfer and real-time synchronization across platforms. - Led the migration from legacy systems to a new application architecture by redesigning database schemas and modifying tables to support new workflows. - Developed backend integrations for enterprise systems to ensure reliable data exchange and processing. - Improved data processing efficiency by optimizing database queries and backend workflows.	
Capgemini Technology Services India Limited <i>Software Engineer</i>	Jun 2019 - Nov 2021
- Developed and deployed high-performance backend APIs, improving employee data accessibility across departments and reducing data retrieval time by 15%. - Designed and optimized PL/SQL stored procedures to support complex data processing and reporting requirements. - Enhanced database performance by improving query execution and optimizing data access patterns. - Automated testing workflows using Selenium and UiPath, improving deployment reliability and reducing manual operational effort.	

PROJECTS

Order Data Lakehouse ETL Architecture (Amazon Internal Project)

2025

- Designed end-to-end data architecture to transfer procurement order data from application databases to a centralized data lakehouse for analytics.
- Built an ETL pipeline to extract and transform application order data and load it into the analytics environment for downstream consumption.
- Configured secure cross-network connectivity using VPC peering between the application VPC and analytics infrastructure to enable controlled data transfer.
- Implemented IAM-based read-only database access and integrated AWS Secrets Manager to securely manage database credentials used by the ETL workflow.
- Automated infrastructure provisioning using AWS CDK to deploy networking, permissions, and supporting resources in a consistent and repeatable manner.
- Enabled operational reporting and business insights by integrating processed datasets with Amazon QuickSight dashboards.

Data Architecture Revamp (Professional Project)

Mar 2022

- Revamped legacy database tables and restructured backend services to support migration from a deprecated data source to a new application architecture.
- Developed new backend APIs to provide structured and normalized datasets, enabling multiple teams to consume data efficiently.
- Redesigned the data processing architecture to incorporate filtering and correction mechanisms, improving overall data quality.
- Reduced operational overhead by 40% by eliminating manual data fixes and improving system reliability.

Seamless Application Integration & Data Transfer (Professional Project)

Mar 2021

- Integrated a new enterprise application into the system architecture to enable seamless data exchange and processing between services.
- Implemented an automated API workflow to fetch, process, and transfer application data into AWS PostgreSQL for reliable storage and retrieval.
- Designed backend workflows to ensure consistent data synchronization across systems.
- Improved system reliability and reduced manual intervention by standardizing API-based data processing pipelines.

BookMyShow-Like Application (Personal Project)

<https://rahulravuri.github.io/Portfolio/BookMyShow.html>

- Designed and implemented backend services for user authentication including secure token-based login and authorization.
- Built core services to manage movie listings, showtimes, and ticket booking workflows.
- Implemented booking logic and database models to support seat allocation and transaction management.
- Structured backend modules to support scalable service communication between booking and movie services.

ChessGame (Personal Project)

<https://rahulravuri.github.io/Portfolio/ChessGame.html>

- Developed a two-player chess game application implementing core game logic including move validation and board state tracking.
- Designed object-oriented models for chess pieces, board states, and game rules.
- Implemented an alert system to notify players when their king is in check.
- Built game state tracking to ensure accurate gameplay progression between moves.